

Network Traffic Analysis Using Wireshark

1. Executive Summary

This project analyzes network traffic within a local network environment using Wireshark. The objective was to observe and understand how devices communicate across the network by capturing and analyzing packet-level data.

Traffic analysis was performed using a Kali Linux virtual machine connected to the local network through a bridged network adapter. During the capture session, several fundamental network protocols were observed, including ARP, DNS, TCP, ICMPv6, and TLS encrypted web traffic.

The analysis demonstrated how devices resolve domain names, establish encrypted web connections, and discover other devices within the local network. No suspicious traffic patterns were identified during this observation exercise.

This project highlights the importance of packet analysis as a foundational skill in cybersecurity, network monitoring, and incident investigation.

2. Scope of Analysis

Category	Description
Analysis Type	Network Traffic Analysis
Environment	Local Home Network
Client System	Kali Linux Virtual Machine
Network Mode	Bridged Adapter
Analysis Tool	Wireshark
Objective	Observe and analyze real network communication

Limitations

- Analysis limited to locally captured network traffic
- No packet injection or attack simulation performed
- Traffic observation conducted under normal browsing activity

3. Methodology

The network traffic analysis followed the following structured steps:

1. Environment Setup
 2. Packet Capture Initialization
 3. Traffic Generation (Web browsing & ping)
 4. Packet Filtering and Protocol Analysis
 5. Identification of Key Network Protocols
 6. Documentation of Observed Traffic Behavior
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4. Environment Setup

The packet capture environment was prepared using a Kali Linux virtual machine running on Oracle VirtualBox.

The virtual machine network adapter was configured in **Bridged Mode**, allowing the system to function as a normal device within the local network.

Network interface verification was performed using the command:

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ip a
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Result

Interface	IP Address	Network
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eth0	192.168.0.27	192.168.0.0/24
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This confirmed that the Kali system successfully joined the local network.

5. Packet Capture Procedure

Wireshark was used to capture live network packets from the active network interface.

The capture process involved the following steps:

1. Launch Wireshark
2. Select the active network interface (eth0)
3. Start packet capture
4. Generate network traffic by visiting websites and performing network tests
5. Stop capture and analyze packets

Traffic filtering was applied to isolate specific protocols for analysis.

6. Protocol Analysis

6.1 ICMP / ICMPv6 Traffic

ICMP packets were observed when performing network connectivity tests using the ping command.

Example command used:

ping google.com

Packet capture revealed **ICMPv6 Echo Request and Echo Reply messages**, indicating that the network prefers IPv6 connectivity when communicating with external hosts.

Observation:

- ICMPv6 Echo Request packets sent from the client system
- ICMPv6 Echo Reply responses received from the remote server
- Successful connectivity verification

123	66.	368979327	DLinkInterna.a8:d9:	Broadcast	ARP	60 who has 192.168.0.47 Tell 192.168.0.1
124	66.	368979665	DLinkInterna.a8:d9:	Broadcast	ARP	60 who has 192.168.0.277 Tell 192.168.0.1
125	66.	368982943	PCSSystemtec.a5:b4:	DLinkInterna.a8:d9:	ARP	42 192.168.0.27 is at 08:00:27:a5:b4:b f
126	66.	7793958191	2001:e68:544d:9839::	2404:6800:a001:818:	ICMPv6_E	118 Echo (ping) request id=0x0002, seq=24, hop Limit=64 (request in 127) (SendTTL=20, Round=21)
127	66.	7793958191	2001:e68:544d:9839::	2404:6800:a001:818:	ICMPv6_E	118 Echo (ping) request id=0x0002, seq=24, hop Limit=119 (request in 126) (SendTTL=20, Round=21)
128	67.	775183642	2001:e68:544d:9839::	2404:6800:a001:818:	ICMPv6_E	118 Echo (ping) request id=0x0002, seq=25, hop Limit=64 (reply in 129) (SendTTL=20, Round=21)
129	67.	784717216	2404:6800:a001:818:	2001:e68:544d:9839::	ICMPv6_E	118 Echo (ping) reply id=0x0002, seq=25, hop Limit=119 (request in 128) (SendTTL=20, Round=21)
130	68.	767676842	2001:e68:544d:9839::	2404:6800:a001:818:	ICMPv6_E	118 Echo (ping) request id=0x0002, seq=26, hop Limit=64 (reply in 131) (SendTTL=20, Round=21)
131	68.	767613470	2404:6800:a001:818:	2001:e68:544d:9839::	ICMPv6_E	118 Echo (ping) reply id=0x0002, seq=26, hop Limit=119 (request in 130) (SendTTL=20, Round=21)
132	68.	787676842	2001:e68:544d:9839::	2404:6800:a001:818:	ICMPv6_E	118 Echo (ping) request id=0x0002, seq=26, hop Limit=64 (reply in 133) (SendTTL=20, Round=21)
133	69.	784364371	2404:6800:a001:818:	2001:e68:544d:9839::	ICMPv6_E	118 Echo (ping) reply id=0x0002, seq=26, hop Limit=119 (request in 132) (SendTTL=20, Round=21)
134	69.	869318509	192.168.0.24	239.255.255.256	SSDP	211 M-SEARCH * HTTP/1.1
135	70.	055346351	DLinkInterna.a8:d9:	Broadcast	ARP	60 who has 192.168.0.47 Tell 192.168.0.1
136	70.	055340749	DLinkInterna.a8:d9:	Broadcast	ARP	60 who has 192.168.0.277 Tell 192.168.0.1
137	70.	055359544	PCSSystemtec.a5:b4:	DLinkInterna.a8:d9:	ARP	42 192.168.0.27 is at 08:00:27:a5:b4:b f
138	70.	055364679	DLinkInterna.a8:d9:	Broadcast	ARP	60 who has 192.168.0.77 Tell 192.168.0.1
139	70.	080336662	2001:e68:544d:9839::	2404:6800:a001:818:	ICMPv6_E	118 Echo (ping) request id=0x0002, seq=28, hop Limit=64 (request in 140) (SendTTL=20, Round=21)
140	70.	080497575	2001:e68:544d:9839::	2404:6800:a001:818:	ICMPv6_E	118 Echo (ping) request id=0x0002, seq=28, hop Limit=119 (request in 139) (SendTTL=20, Round=21)
141	70.	781915752	192.168.0.24	239.255.255.256	SSDP	211 M-SEARCH * HTTP/1.1
142	71.	783173070	2001:e68:544d:9839::	2404:6800:a001:818:	ICMPv6_E	118 Echo (ping) request id=0x0002, seq=29, hop Limit=64 (reply in 143) (SendTTL=20, Round=21)
143	71.	791991448	2404:6800:a001:818:	2001:e68:544d:9839::	ICMPv6_E	118 Echo (ping) reply id=0x0002, seq=29, hop Limit=119 (request in 142) (SendTTL=20, Round=21)
144	71.	792782304	192.168.0.24	239.255.255.256	SSDP	211 M-SEARCH * HTTP/1.1


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ame 121: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on Interface eth0, id 0   0000    a0 a3 fa 8 d c 6 00 27 a5 b4 bf 86 dd 60 6c 
 hernet II, Src: PCSSystemtec.a5:b4:bf (08:00:27:a5:b4:bf), Dst: DLinkInterna.a8:d9:c (0a:03:f0:a8:d9)  0000    75 b5 00 40 3a 20 01 e6 84 d4 98 39 67 7b u : l t

```

6.2 DNS Traffic

Domain Name System (DNS) packets were captured when the system attempted to resolve domain names during web browsing activity.

Example packet observation:

Standard query AAAA safebrowsing.googleapis.com

This indicates that the system requested the IPv6 address of Google's Safe Browsing service.

The DNS response returned an IPv6 address for the requested domain.

Observation:

- DNS query sent from client (192.168.0.27) to local gateway (192.168.0.1)
- DNS resolver returned the IPv6 address associated with the requested domain
- Successful domain name resolution

234	14.079440990	192.168.0.1	192.168.0.27	DNS	100 Standard query response 0xc000 AAAA firefox-sellings-atlascrystals.com.mozilla.net CNAME mozilla.mop.fastly.net AAAA 2004:4042::f0
255	14.679446377	192.168.0.1	192.168.0.27	DNS	107 Standard query response 0x3482 AAAA ads-img.mozilla.org AAAA 2a04:4e42::347
473	41.505224206	192.168.0.27	192.168.0.1	DNS	87 Standard query 0x129a A safebrowsing.googleapis.com
474	41.505748020	192.168.0.27	192.168.0.1	DNS	87 Standard query 0xe19b AAAA safebrowsing.googleapis.com
475	41.512187197	192.168.0.1	192.168.0.27	DNS	103 Standard query response 0x129a A safebrowsing.googleapis.com A 172.217.25.10
476	41.512187508	192.168.0.1	192.168.0.27	DNS	115 Standard query response 0xe19b AAAA safebrowsing.googleapis.com AAAA 2404:6800:4001:80c::200a
491	41.545590302	192.168.0.27	192.168.0.1	DNS	70 Standard query 0xb7e5 A o.pki.goog
492	41.545646417	192.168.0.27	192.168.0.1	DNS	70 Standard query 0xaae2 AAAA o.pki.goog
493	41.552020122	192.168.0.1	192.168.0.27	DNS	121 Standard query response 0xb7e5 A o.pki.goog CNAME pki-goog.l.google.com A 172.217.25.67
494	41.552820252	192.168.0.1	192.168.0.27	DNS	133 Standard query response 0xaae2 AAAA o.pki.goog CNAME pki-goog.l.google.com AAAA 2404:6800:4001:811::2003
1258	48.465838383	192.168.0.27	192.168.0.1	DNS	74 Standard query 0xd47f A www.google.com
1259	48.466819613	192.168.0.27	192.168.0.1	DNS	74 Standard query 0x3f7d AAAA www.google.com
1263	48.493771605	192.168.0.1	192.168.0.27	DNS	202 Standard query response 0xd47f A www.google.com AAAA 2001:4860:482a:7700:: AAAA 2001:4860:4828:7700:: AAAA 2001:4860:482c:77
1264	48.493771979	192.168.0.1	192.168.0.27	DNS	298 Standard query response 0x3f7d AAAA www.google.com A 142.251.150.119 A 142.251.157.119 A 142.251.154.119 A 142.251.151.119 A 142.2
1284	48.534208802	192.168.0.27	192.168.0.1	DNS	70 Standard query 0x72da A o.pki.goog
1287	48.558959662	192.168.0.1	192.168.0.27	DNS	121 Standard query response 0x72da A o.pki.goog CNAME pki-goog.l.google.com A 172.217.25.67
1288	48.559081633	192.168.0.27	192.168.0.1	DNS	70 Standard query 0x9d08 AAAA o.pki.goog

6.3 HTTPS and TLS Traffic

Encrypted web communication was observed when accessing websites through the browser.

Packet analysis identified TCP communication over **port 443**, which is the standard port used for HTTPS traffic.

Captured packets showed:

TLSv1.3 Application Data

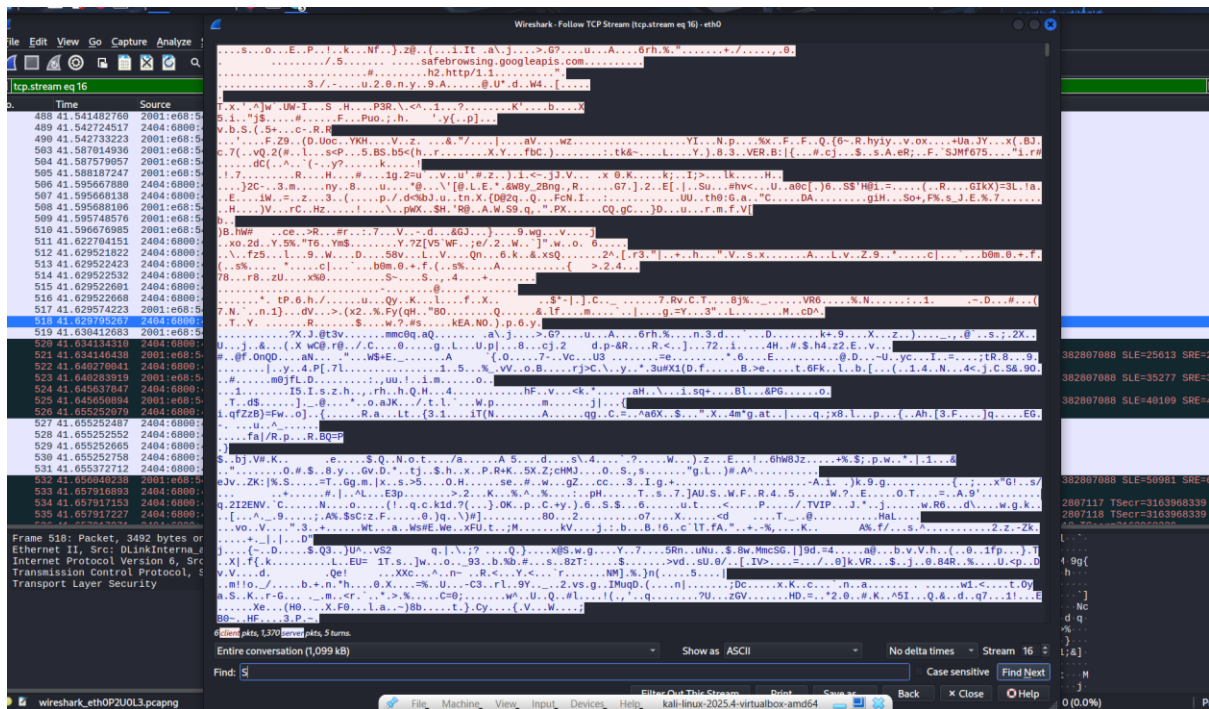
This confirms that the browser established a secure encrypted connection with the remote web server.

Observation:

- TCP connection established between client and remote server
- HTTPS communication conducted over TCP port 443
- TLSv1.3 encryption applied to protect web traffic

Due to encryption, the content of the web traffic could not be directly viewed within Wireshark.

514	41.629522532	2404:6800:4001:80c::	2001:e68:544d:9839::	TLSv1.3	2502 Application Data, Application Data
515	41.629522601	2404:6800:4001:80c::	2001:e68:544d:9839::	TLSv1.3	3710 Application Data, Application Data, Application Data
516	41.629522668	2404:6800:4001:80c::	2001:e68:544d:9839::	TLSv1.3	3710 Application Data, Application Data, Application Data
517	41.629574223	2001:e68:544d:9839::	2404:6800:4001:80c::	TCP	86 41000 → 443 [ACK] Seq=2586 Ack=7711 Win=71680 Len=0 TSval=3163968335 TSecr=3382807088
518	41.629795267	2404:6800:4001:80c::	2001:e68:544d:9839::	TLSv1.3	3492 Application Data, Application Data, Application Data
519	41.630412683	2001:e68:544d:9839::	2404:6800:4001:80c::	TCP	86 41000 → 443 [ACK] Seq=2586 Ack=21989 Win=58880 Len=0 TSval=3163968336 TSecr=3382807088
520	41.634134310	2404:6800:4001:80c::	2001:e68:544d:9839::	TLSv1.3	1294 [TCP Previous segment not captured] , Application Data
521	41.634146438	2001:e68:544d:9839::	2404:6800:4001:80c::	TCP	98 [TCP Dup ACK 519#1] 41000 → 443 [ACK] Seq=2586 Ack=21989 Win=71680 Len=0 TSval=3163968339
522	41.640270041	2404:6800:4001:80c::	2001:e68:544d:9839::	TLSv1.3	1294 [TCP Previous segment not captured] , Application Data
523	41.640283919	2001:e68:544d:9839::	2404:6800:4001:80c::	TCP	106 [TCP Dup ACK 519#2] 41000 → 443 [ACK] Seq=2586 Ack=21989 Win=71680 Len=0 TSval=3163968346
524	41.645637847	2404:6800:4001:80c::	2001:e68:544d:9839::	TLSv1.3	1294 [TCP Previous segment not captured] , Continuation Data



6.4 ARP Traffic

Address Resolution Protocol (ARP) packets were observed during communication between devices within the local network.

Example captured packet:

Who has 192.168.0.24? Tell 192.168.0.1

This indicates that the network gateway attempted to identify the device associated with the specified IP address.

ARP is used to resolve IP addresses into MAC addresses, enabling Ethernet communication within the local network.

Observation:

- ARP broadcast requests issued by the network gateway
- Devices within the network respond with their MAC address
- Enables communication between devices within the same subnet

439	23.256552352	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.7? Tell 192.168.0.1
444	26.945194084	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.24? Tell 192.168.0.1
445	29.609227248	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.4? Tell 192.168.0.1
446	29.609227942	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.24? Tell 192.168.0.1
447	32.273088563	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.4? Tell 192.168.0.1
448	32.273088971	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.24? Tell 192.168.0.1
457	35.859050553	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.4? Tell 192.168.0.1
458	35.859050877	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.24? Tell 192.168.0.1
461	38.521280676	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.4? Tell 192.168.0.1
471	41.185128566	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.4? Tell 192.168.0.1
472	41.185128915	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.24? Tell 192.168.0.1
997	44.867421711	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.4? Tell 192.168.0.1
998	44.867422109	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.24? Tell 192.168.0.1
1474	50.196653407	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.4? Tell 192.168.0.1
1475	50.196653622	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.24? Tell 192.168.0.1
1809	53.477421091	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.27? Tell 192.168.0.1
1810	53.477431767	PCSSystemtec_a5:b4:...	DLinkInterna_a8:d9:...	ARP	42 192.168.0.27 is at 08:00:27:a5:b4:bf
1852	53.790663446	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.4? Tell 192.168.0.1
1853	53.790663765	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.24? Tell 192.168.0.1
2135	56.443516568	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.4? Tell 192.168.0.1
3890	59.212957711	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.4? Tell 192.168.0.1
3891	59.212958163	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.24? Tell 192.168.0.1
3892	59.212958194	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.27? Tell 192.168.0.1
3893	59.212975802	PCSSystemtec_a5:b4:...	DLinkInterna_a8:d9:...	ARP	42 192.168.0.27 is at 08:00:27:a5:b4:bf
4865	62.905734791	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.4? Tell 192.168.0.1
4866	62.905735234	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.24? Tell 192.168.0.1
4867	62.905735265	DLinkInterna_a8:d9:...	Broadcast	ARP	60 Who has 192.168.0.27? Tell 192.168.0.1
4868	62.905776863	PCSSystemtec_a5:b4:...	DLinkInterna_a8:d9:...	ARP	42 192.168.0.27 is at 08:00:27:a5:b4:bf

7. Key Observations

The packet capture revealed several normal network behaviors:

- DNS queries used to resolve domain names before establishing connections
- HTTPS communication encrypted using TLSv1.3
- ARP broadcasts used to identify devices within the local network
- ICMPv6 traffic used for connectivity testing

All observed traffic patterns were consistent with normal network activity.

8. Security Insights

Although no malicious activity was detected during this analysis, several security insights can be derived:

- Encrypted HTTPS traffic prevents packet inspection of sensitive data
- DNS resolution is a critical step in web communication
- ARP traffic can be abused in attacks such as ARP spoofing or man-in-the-middle attacks
- Packet analysis can be used by security analysts to detect abnormal network behavior

Understanding these protocols is essential for network monitoring and incident response.

9. Conclusion

This project demonstrated how network traffic can be captured and analyzed using Wireshark.

Through packet inspection, several key protocols were identified, including ARP, DNS, ICMPv6, TCP, and TLS encrypted HTTPS communication.

The analysis confirmed that the observed traffic represented normal network activity and illustrated the fundamental processes involved in web communication and local network device discovery.

Packet analysis provides valuable insight into network behavior and serves as a critical skill for cybersecurity professionals involved in network monitoring, threat detection, and digital forensics.